

## DAEN 427 Homework 1

Due date: Aug 26, 2025

1. Explain the difference between a *confounder* and a *collider* in the context of a DAG. Why is it important not to condition on a collider when trying to estimate a causal effect?

**Answer:**

2. You are studying whether a new training program ( $A$ ) improves job performance ( $Y$ ). Based on domain knowledge, age and prior experience influence both program participation and job performance. Gender is believed to influence only program participation. Construct a DAG using this information, and identify the minimum set of covariates you would need to adjust for to estimate the causal effect of  $A$  on  $Y$ .

**Answer:**

3. Refer to the “Uncomplicated Appendicitis” DAG example in the slides. Discuss the role of *income* and *primary language* in this model. Which one is unmeasured? What potential biases could arise from not including it in your analysis?

**Answer:**

4. Consider a DOG where a treatment variable  $A$  causally affects an outcome  $Y$ , and there exists a confounder  $C$  that influences both  $A$  and  $Y$ .
- (a) Explain why the conditional probability  $P(Y | A)$  is not equal to the causal effect  $P(Y | \text{do}(A))$ .
  - (b) Describe how adjusting for  $C$  allows us to estimate the causal effect using observational data.

**Answer:**